Workshop: solar thermal energy

Cooking, preservation, observing heat and an experiment worksheet.

90 min + solar observation

Connect cooking knowledge with solar heat. Compare parabolic cookers, box ovens and dryers, and decide together what is worth trying.

Before starting

Agree voluntary participation, language, timing and what will be recorded. Use the priority chosen by the group. Durations are indicative. Close by returning results and agreeing follow-up.

01 / Start with everyday cooking

15 min. Discuss foods, timing, fuels, seasons and care work. Ask which practice could be complemented by solar energy, respecting the group's choices.

Food cards and an activity map.

02 / Meet three technologies

20 min. Use the downloadable guide to compare concentration in a parabola, heat retention in a box oven and airflow in a dryer. Rank advantages and limits for the local context.

Downloadable guide, cardboard and drawings.

03 / Observe and experiment

30 min. With a facilitator, compare two water containers under controlled conditions and record time, temperature and cloud cover. Use a safe solar device; if sunlight is unavailable, build a non-concentrating model.

Suitable containers, water, thermometer, timer, gloves and heat protection. Avoid looking at concentrated reflections.

04 / Interpret without overpromising

15 min. Compare records. What changed with orientation, clouds or wind? Cooking time depends on weather, equipment and food. An educational prototype does not establish food-safe cooking.

Table: time, temperature, conditions and observations.

05 / Share a repeatable learning

10 min. Choose a use to explore next, record local climate adaptations and explain the thermal principle in your own words. Agree who looks after the materials and when to review them.

Suggested evidence: object photographs, temperature records and an adaptation sheet.

WORKSHEET

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Choose what to observe

Compare two conditions keeping container, water volume and observation time equal. Record what changes and what is not controlled. Do not taste food during a trial that does not establish safe cooking.

Device	Principle	What to review
Parabola	Concentrates light	Orientation, reflections and hot focus
Box oven	Retains heat	Insulation, cover and temperature
Solar dryer	Heat and airflow	Moisture, hygiene and suitable process

Observation record

Time	Temp A	Temp B	Cloud / wind / notes
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Do not look at concentrated reflections or leave a concentrator unattended. Use gloves and a marked area. If conditions are unsafe, work with a non-concentrating model.

Genially / solar thermal energy (Spanish)

Close and agree

What can we explain in our own words? What would we adapt? What needs confirmation? Keep an authorised record, a responsible role and a next review.